

Adaptive multihop broadcast protocols for ad hoc networks

Andras Kokuti, Vilmos Simon

Abstract—Abstract— In line with traditional communication systems, more and more attention is given to autonomous, self-organized networks with no central infrastructure, based on peer-to-peer communication, like mobile ad hoc networks. Designing multihop broadcast protocols for ad hoc networks is a complex problem as the task of these protocols is to disseminate messages in a network effectively while avoiding unnecessary use of resources. To solve this problem, this paper presents three adaptive multihop broadcast protocols, which assign a broadcasting probabilities to the mobile nodes, determining them from the network parameters (like the degrees of the nodes, distance of the nodes from each other). The presented novel 3-phase handshake gossiping protocols, the DBHG, the VBHG and the AVBHG generate significantly less duplicates while giving a faster coverage of the nodes than the other examined protocols.

Keywords—adaptive protocols, multihop broadcast, mobile ad hoc networks

I. INTRODUCTION

TODAY more and more are appreciated networks with no infrastructure, which are self-organized and where the communication is based on a peer-to-peer model. A major representative of these networks is the mobile ad hoc network (MANET[1]), which is a self-configuring network, consisting of mobile devices, that can communicate with each other via a radio interface. The distributed and self-configuring nature of these networks, combined with the ease and flexible deployment of such networks, make MANETs appealing for a wide range of application scenarios including, e.g., emergency situations, sensor networks for environmental monitoring[2], vehicular ad hoc networks[3], and many others[4], [5].

The central problem of multihop broadcast protocols for MANETs is to decide when and who should retransmit messages. Nodes have to forward packets so the message reaches every part of the network, however the performance relies heavily on the set of nodes that do this. When nodes decide whether to retransmit or not, they actually decide if they are part of the forwarding set or not. Too many retransmissions cause collisions and waste the network bandwidth, but choosing the smallest forwarding set is not easy because a global view of the network is not available, and local information gets obsolete very quickly if the velocity of nodes is high. There is also a risk if the number of forwarding nodes is too small, because then the message may not reach every node.

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Andras Kokuti and Vilmos Simon are with the Department of Telecommunication, Budapest University of Technology and Economics, Budapest (e-mail: svilmos@hit.bme.hu).

The naive first implementation was the blind flooding[6] algorithm. The operation of BF algorithm is very simple, once a node receive a message, immediately rebroadcasts it. We might think this is a good solution for data distribution, because it is very simple, and covers all the nodes (assuming that there are no separation), but it was realized very soon, that this is very far from optimal. The effect of frequent packet repetition may occur collisions in the medium and can lead to serious congestions, as broadcast storms[7].

The Counter Based Method, originally introduced in [8] is one of the first controlled broadcast methods. It is based on a simple observation, that if a duplicate of a packet is received, then the probability of reaching any new node is low. To exploit this idea, the nodes do not immediately transmit on the receipt of a packet, but instead they wait for a random time, which is called Random Assessment Delay (RAD). The original method has different adaptive versions [9], which try to adapt the length of the RAD and the threshold of the duplicate counter to the current network conditions.

Another broadcast method is the Gossiping algorithm [8], where every node broadcasts the heard message with a pre-defined probability. The optimal probability can be calculated off-line, or can be learned adaptively. Some of these adaptive versions are covered in [10].

Hypergossiping [11] is specifically designed for partitioned networks, where nodes are mobile, and partitions join and split from time to time. It is an advanced version of the Gossiping algorithm, extended by neighbor information and partition join detection.

A fine example of a self-pruning algorithms is the Scalable Broadcast Algorithm (SBA) [12]. It requires a 2-hop neighbor information and the last sender ID in the broadcast packet.

A quite different approach from the algorithms discussed so far is the IOBIO algorithm[13]. It is a variation of the SPIN [14] dissemination protocol. The goal of the protocol is to reduce the unnecessary load of neighboring nodes by duplicate or unneeded data (spamming).

In Section II we introduce our novel adaptive multihop broadcast protocols, while the performance of these protocols are presented in Section III. Finally, Section IV provides the conclusion.

II. NOVEL ADAPTIVE MULTI HOP BROADCAST PROTOCOLS

A. Three-way handshaking

As we have seen in the previous chapter, there are many protocols existing already for multihop broadcasting, but we could not claim for any solution, that it can perform optimally

in every kind of environment. Because of the the specificities of the mobile ad hoc networks, where every network element can move with random direction and velocity, we can not rely on any global information. The protocols must make right decisions from their own local information. Our new protocols try to collect some local information, like the degrees of the nodes, distances of the nodes from each other, contributing to a more efficient solution then the similar protocols found in the literature. In this subsection the common properties of the three novel protocols are described, while the following subsections describe them more thoroughly.

Beyond data message distribution, there is a signal message based communication and an adaptive forwarding probability determination too in the protocols. The protocols differs from each other only in the methods of calculating the message forwarding probabilities, as it will be discussed in the subsection of the given protocol. We have assumed that the devices have the knowledge of their actual geographical positions (with the help of GPS transceivers or through other positioning techniques, which is a realistic assumption in today's smart phones and notebooks) . All the three protocols consists from the the same phases.

The first phase is called RTB[15] (Ready-To-Broadcast), which is similar to the RTS phase in CSMA/CA[16]. The device, which has got a data message to disseminate, broadcasts an RTB signal message in its radio range. It places in the packet its own unique identifier (for example MAC address) and in a multi-message communication (where mobile nodes disseminate different kind of data messages) the message type and identifier.

In Figure 1 the black point marks the sender node, the grays are the nodes in the senders range, and the whites show the other nearby devices. The transmitting device, marked with black colour, is entering the first phase by disseminating an RTB packet. The neighboring nodes (gray colour) in its radio range receive the message due to the broadcast.

In the next phase all of the nodes, which received an RTB message, start a random timer from the $[0; t_{CTB}]$ range. This timer is responsible for ensuring that the returned CTB (Clear-To-Broadcast) messages do not cause broadcast storms, and to avoid collisions. The message includes the CTB sender's and the "recipient"'s (which sent the RTB packet int the previous phase) unique identifier. With the returned identifiers, the central (marked with black colour in Figure 1) node can decide,

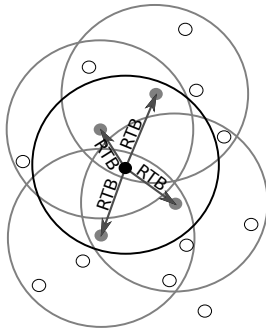


Fig. 1. The first phase

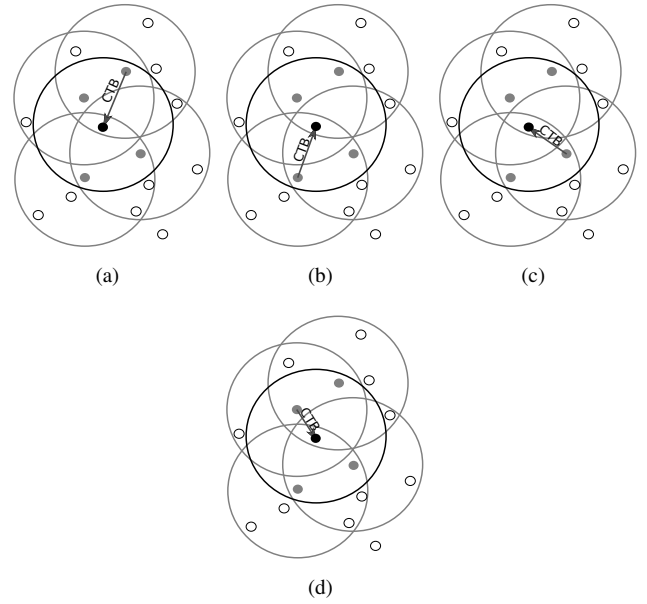


Fig. 2. The second phase

if it is the recipient of the packet or another 2-hop distance node. The message also contains the geographical coordinates of the CTB sender (by using the GPS transceiver or other positioning technique) and the number of its neighbors, namely its degree. To detect their neighbors, the nodes periodically distribute HELLO and HELLO-REPLY signal messages. All devices transmit a HELLO message at appropriate intervals for monitoring their own environment. If HELLO-REPLY is received, the node increases the number of its neighbors, and if HELLO message is received, it answers with a HELLO-REPLY message. So the received HELLO-REPLY messages help to determine the actual degree of the given node. In addition to these, CTB messages also include the identifiers of the messages (getting it from the RTB packet), which are needed by the CTB sender node.

First, as shown in Figure 2, the node in the (a) subfigure gets the opportunity to broadcast the CTB message, because its timer expired earlier than for the other nodes which also started a timer upon receiving the RTB message. Then nodes, shown in (b), (c) and (d) subfigure will transmit one after other, determined by their timers . Meanwhile the central node is waiting for a t_{DATA} time, which is calculated as the sum of the t_{CTB} and the maximum propagation time.

The last phase is responsible for the data message dissemination, the first two phases use only signal messages. The central node starts the data message transmission (which sent out previously the RTB packet) after its t_{DATA} timer expired. First, it summarizes based on its 1-hop neighbor list, which nodes are currently within its radio range, and to these IDs (which can be found in the reply CTB messages) assigns the degrees and the coordinates. After it calculates the retransmission probability for each adjacent device. From the CTB messages the central node can determine which data messages need to be disseminated, by calculating the union of the requested data messages.

Figure 3 shows an example how is the central node de-

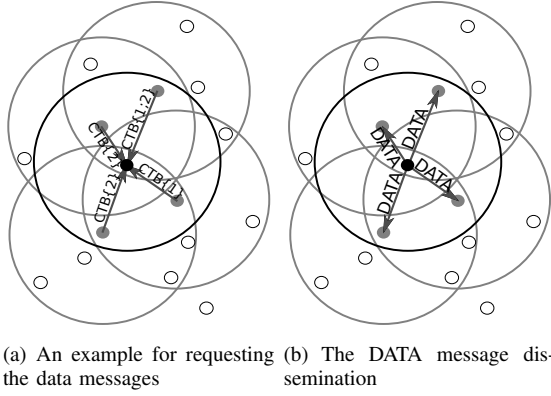


Fig. 3. The third phase

termining which data messages need to be disseminated. The adjacent devices, marked by gray colour, put the requested data message IDs in the CTB packets (in braces). Based on these identifiers, the central node calculates which messages (here, the messages with ID 1 and 2), need to be broadcasted. Thus, all parameters are known (messages, probabilities) in order to complete the information dissemination.

The DATA message's header contains the sender (same as RTB sender) unique identifier and the calculated message forwarding probabilities, the payload is comprised of the requested data messages. The sender's identifier is necessary as there may be another 2-hop distance nodes, which are in the same transmission phase, therefore the common neighboring nodes can distinguish which node's DATA packet have been received. From the rest of the header every adjacent node get its message forwarding probability, as it is assigned to the nodes ID

All the three multihop broadcast protocols have a single message[17] and a multi message version too[18]. As we mentioned, the protocols only differ in their methods of the probability calculation, so in the next subsections the concrete protocols will be described.

B. Distance Based Handshake Gossiping (DBHG)

For the Distance Based Handshake Gossiping protocol the degree of the nodes is not needed, so that can be omitted from the CTB message. Moreover, the periodically sent topology discovery signal messages (HELLO, HELLO-REPLY) are unnecessary. For calculating the probabilities only the geographical coordinates are needed, which can be found in the CTB message. The calculation is based on the nodes distances, hence the name, which is defined using the following formula.

$$d_i = \sqrt{(\Delta x)^2 + (\Delta y)^2 + (\Delta z)^2} \quad (1)$$

$$d_i = \sqrt{(x_0 - x_i)^2 + (y_0 - y_i)^2 + (z_0 - z_i)^2} \quad (2)$$

The x_0, y_0, z_0 parameters are the central node's, while x_i, y_i, z_i are the i . adjacent device's coordinates. The central node selects the largest distance, ie the highest value, it will be d_{max} .

$$d_{max} = \max_{\forall i} \{d_i\} \quad (3)$$

The protocol, based on these parameters, assigns the following message forwarding probability to the i . neighboring node:

$$p_i = \frac{d_i}{d_{max}} \quad (4)$$

This means, as being farther from the central node, the message forwarding probability will get higher for the given adjacent node. This way the duplication caused by closer nodes can be avoided, moreover the area covered by more distant nodes is more likely to contain previously not covered nodes[19]. This means that the furthest node's (which is d_{max} away from the central node) associated probability variable will be 1, so it will initiate the retransmission with 100%. Thus in the set of the neighboring nodes there will be at least one node (the furthest), which will continue the information dissemination.

C. Valency Based Handshake Gossiping (VBHG)

A shortcoming of the previous protocol is that it is not considering the degree of the nodes when assigning the message forwarding probabilities to the neighboring nodes. It may happen, that the furthest adjacent device has no other neighbors, than the initial transmitter, nevertheless it has got the 100% forwarding probability because of its position. Furthermore the protocol still not "penalize" enough the nearby nodes. In order to avoid unnecessary duplication of messages, the nearby nodes with a high degree need to get a low chance for the retransmission, while the farther nodes should have a higher probability. This version of the protocol is trying to achieve this goal. Of course, in this case it is necessary to distribute periodically the topology discovery signal messages, and some additional parameters are needed. The d_i, d_{max} variables are identical to those used in the DBHG protocol. The $d_{average}$ indicates the average value of the distances, calculated as an arithmetic mean of the central node-neighbors distances

$$d_{average} = \frac{1}{n} * \sum_{i=1}^n d_i \quad (5)$$

where n is the number of adjacent nodes, while f_i is the i . neighboring node's degree. The neighbors can be distinguished by the unique identifier in the CTB message. Similar to d_{max} an f_{max} is needed for VBHG, to determinate the probabilities.

$$f_{max} = \max_{\forall i} \{f_i\} \quad (6)$$

The probability is calculated the following way for the VBHG protocol:

$$p_i = \begin{cases} \frac{d_i}{d_{max}} - \frac{d_i}{d_{max}} * \frac{f_i}{f_{max}}, & \text{if } d_i < d_{average} \\ \frac{d_i}{d_{max}} + \frac{d_{max}-d_i}{d_{max}} * \frac{f_i}{f_{max}}, & \text{if } d_i \geq d_{average} \end{cases} \quad (7)$$

As equation 7 shows the calculation is comprised from two disjoint cases. The upper equation is used, when the adjacent device is closer to the transmitter than the average distance. In this case, the goal is to assign a lower retransmission probability, because this node's radio range is already largely covered by the central node, so the rebroadcast would cause many unnecessary duplications. On the other hand, when the device is far from the transmitter, more likely it covers new, still not covered nodes, the lower equation is utilized for this case.

D. Average Valency Based Handshake Gossiping

The third version of the protocol is based on the experience provided by the two previous versions. It uses one more parameter, the $f_{average}$ indicates the average valency of the network:

$$f_{average} = \frac{1}{n} * \sum_{i=1}^n f_i \quad (8)$$

where n is the number of the neighbors for the given node. Consequently, the probability is determined by :

$$p_i = \begin{cases} \frac{d_i}{d_{max}} + \frac{d_{max}-d_i}{d_{max}} * \frac{f_i}{f_{max}} & \text{if } f_i \geq f_{average} \\ & \text{and } d_i \geq d_{average} \\ \frac{d_i}{d_{max}} + \frac{d_{max}-d_i}{d_{max}} * \frac{f_i}{f_{max}} & \text{if } f_i \geq f_{average} \\ & \text{and } d_i < d_{average} \\ \frac{d_i}{d_{max}} & \text{else} \end{cases} \quad (9)$$

As it can be seen, the probability calculation is a combination of the previous two protocol probability assignments, therefore those nodes get lower probabilities for retransmission which are closer to the central node than the average distance, and in the same time they have a high valency value.

III. PERFORMANCE ANALYSIS

A. Simulator

For measuring the performance of the protocols a simulator [20] was implemented in C++, in which the protocols can operate under similar conditions as in a real MANET.

A modified version of the random waypoint mobility model [21] was used, where the waypoint should be chosen in the area of node's movement range. When the mm parameter is on (a boolean variable is true) multi message communication is simulated while in false state - default - there is only one data message disseminated in the network.

TABLE I
THE PARAMETER LIST

Parameters name	Description
N[db]	Number of the nodes
R[e]	Transmission range
R'[e]	Movement range
T[e ²]	Simulation area
Pid	Simulated protocol's ID
t[db]	Number of tests
mm	Single/multi message communication

Table 1 *Pid* shows the implemented protocols, our protocols (DBHG, VBHG and AVBHG) were compared to Gossiping, Adaptive Gossiping and Modified Adaptive Gossiping, introduced in Section 1. In the Gossiping protocol the retransmission probability can be adjusted (in our measurements it was set to 0.7), while the retransmission probabilities of the Adaptive Gossiping protocol were depending on the nodes valency values. The third one is a modified version of the Adaptive Gossiping, where transmission phases are used, identically to the ones used in our protocols.

B. The performance indicators

One of the most important indicators in data dissemination is the number of the duplications, as it effects the performance of the whole network (speed of the data dissemination, energy consumption). The duplication specifies how many unnecessarily received messages exist in the system (the nodes already own them, so resources are consumed to spam other nodes). It can be calculated as a difference between the requested and the received messages in the system:

$$duplication = \sum_{i=1}^N i_{message_{received}} - \sum_{i=1}^N i_{message_{requested}} \quad (10)$$

The high number of duplications in the system would mean inefficient utilization of the resources, so optimally this metric would be 0, that means all the devices have received only once the message, which were needed by them.

Another important performance metric is the coverage, which indicates what percentage of the nodes requests are completed:

$$coverage[\%] = \frac{\sum_{i=1}^N i_{message_{received'}}}{\sum_{i=1}^N i_{message_{requested}}} * 100\% \quad (11)$$

where $message_{received'} \neq message_{received}$, as in this case the number of duplications is not considered, only if a node have received the requested message. In a single message communication it is equal with the number of the nodes, which received the message. One of the goals of the protocols is to get a high coverage in the shortest time possible. As it can be seen, the above presented two indicators are contradicting each other. For example, the BF protocol delivers the fastest coverage, but it can not be said it is a good solution, as it is extremely wasteful, causing too much collisions and duplications in the system. Therefore a trade-off must be found between the coverage speed and the efficiency in means of duplications.

The third indicator is the number of sent messages, it shows how many broadcast disseminations occurred in the network. It can be calculated as the sum of the nodes broadcast transmissions.

The fourth parameter shows the time needed for the given protocol to achieve a full or prescribed level of coverage of the nodes, for example how much time does it needs to

disseminate a given message for a given percentage or for the whole population of the nodes.

C. Evaluation of the results

The protocols performance was evaluated by the help of the above described parameters fixing the coverage level on 95%. Thus, the simulation will stop when the specified protocol reaches 95% coverage.

1) *Single message scenario*: Consider first the single message communication.

When there is only one type of data message to be disseminated in the system, we have assumed that all of the network devices need to receive it. The single message scenario parameters are presented in Table II. It can be seen from this table that simulations were run for two different sized networks, one which consists of 50 nodes, while the other one of 150 nodes. In order to reduce the simulator's standard deviation, the result were calculated by taking the average of 100 tests.

The results show (Table III), that the protocols, which use the 3-phase handshaking (RTB, CTB and DATA message exchange), performed almost identically when considering the number of duplications. However it should be noticed that these produce much less duplicates, than the two other protocols (Gossiping and Adaptive Gossiping). This is because of the 3-phase handshaking, which, if the neighboring devices are equipped with the data message, does not allow the unnecessary retransmission. This also explains the differences in the number of the sent messages. When examining the coverage time, the situation is somewhat different. The DBHG, AVBHG and Gossiping protocol performed better 20-30% than the other three. The VBHG from this aspect is behind the other versions, because the devices within the average distance are highly penalized, thereby decreasing the opportunity to disseminate messages. But at the other hand in a denser environment, it can reduce the number of duplications. In the single message scenario by using signal messages (and the 3-phase handshaking) we were able to drastically reduce the

TABLE II
SINGLE MESSAGE SCENARIO

Parameters name	Value
N[db]	50,150
R[e]	0,2e
R'[e]	0,1e
T[e ²]	4e ²
Pid	1,2,3,4,5,6
t[db]	100
mm	false

TABLE III
SINGLE MESSAGE SCENARIO RESULTS I.

Protocol	Duplications	Sent messages	Coverage time
DBHG	98,21	44,15	52,62
VBHG	96,35	43,32	66,73
AVBHG	97,51	43,63	55,85
Gossiping	1631,98	1044,63	56,01
Adap. Gos.	1627,57	810,53	77,77
Mod. Adap. Gos.	97,54	42,32	70,95

TABLE IV
SINGLE MESSAGE SCENARIO RESULTS II.

Protocol	Duplications	Sent messages	Coverage time
DBHG	496,52	104,73	4,53
VBHG	447,64	95,83	7,2
AVBHG	482,55	102,25	5,3
Gossiping	2231,08	483,89	6,83
Adap. Gos.	2896,74	564,07	6,36
Mod. Adap. Gos.	571,76	102,6	6,25

number of sent messages, and this way also the number of duplications.

In a case of a denser network (when having 150 nodes) the VBHG generates the least duplications, by 10% less than the other handshake based Gossiping protocols (Table IV). Respectively the Modified Adaptive Gossiping's (which also uses the CTB, RTB and DATA messages), did not deliver a good performance as the other 3-phase based protocols. This is because its probability function is not so complex (it only uses the valency value) as for the others. Furthermore the DBHG spreads the data message the fastest, which is the positive impact of the distance-based($\frac{d_i}{d_{max}}$) probability assignment, while its weak point is that it generates more duplicates than the other two novel protocols.

In the single message scenario our protocols are much more effective than the Gossiping and Adaptive Gossiping. They duplicate significantly less, while the DBHG can produce a faster coverage. All three versions of our protocols should be installed on the device, as they differ only in the way of calculating the probability, which does not cause a large extra cost. Thus, the nodes can adaptively change the function of the probability assignment, depending on which parameter(s) needed to be optimized. If the system requires a rapid dissemination the device would select the DBHG version, however, if there is a high cost to deliver a message, then they should switch to VBHG, while AVBHG would be a trade-off between the two aspects.

2) *Multi message*: In the case of a multi message communication several type of different data messages exist in the network. We assume that the devices can be divided into sets depending on which messages are they interested in. Therefore for the full coverage it is not needed to deliver all the messages for all the devices. We assumed that there are three types of data message in the system. If there are N nodes in the system, they are distributed to seven $\approx \lceil \frac{N}{7} \rceil$ pieces sets, in which the nodes have the same interest for messages. Thus, the node sets can be mapped to the subsets of the data messages. For example, we can say that, the second set of nodes needs the first and second type of messages, while the sixth one only requires the second one. The elements of the given set have been randomly selected to avoid the message grouping.

The simulator input parameters are very similar to the single message scenario (Table II), but for the multi message communication, only results for the denser network (consisting of 150 nodes) were presented due to the lack of space, as this scenario is more representative for the multi message scenario.

As it can be seen (Table V), when observing the number of duplications and sent messages, the results are similar to the

TABLE V
MULTI MESSAGE SCENARIO RESULTS

Protocol	Duplications	Sent messages	Coverage time
DBHG	1233,18	205,66	11,55
VBHG	1111,43	177,71	23,5
AVBHG	1217,53	204,22	11,44
Gossiping	6895,99	498,6	7,04
Adap. Gos.	8381,62	582,33	6,1
Mod. Adap. Gos.	1440,29	226,35	11,92

previous single message case. However it can be seen, that in the multi message scenario the protocols, which use 3-phase handshaking, are slower than the others (Gossiping, Adaptive Gossiping). It is because in the second (CTB) phase the nodes send back to the transmitter a message, which indicates their requirements. Therefore it can possibly happen, that not all of the data messages are sent by the actual central node, while with the Gossiping it can not occur. Furthermore it can also happen, that a node does not possess the data message, which is required by an adjacent device, because it is not needed for the central node, which can not occur when using Gossiping. In the single message case there was only one type of data message, which was needed by every node.

In the multiple message communication case our protocols are still delivering better results as they duplicate significantly less, and can reduce the unnecessary message forwarding too. These results were achieved because of the 3-phase handshaking mechanism. However for a multiple message communication it can also be a drawback, as the protocols, which do not use these phases, provide a better performance when the coverage speed is an important aspect.

IV. CONCLUSION

In the literature there are various protocols for multihop broadcasting in MANETs, starting with the simplest BF protocol to the more complex IOBIO, which uses several signal messages to reduce the number of duplications. Every protocol have some negative features, because of the dynamically changing MANET environment. In this network a trade-off should be achieved between the dissemination speed, simplicity and efficiency, because the protocols can not rely on global information, and due to the mobility of the nodes the local information becomes obsolete fast. Thus, in this environment the protocols must distribute signal messages to get some additional information, nevertheless that these messages are overhead for the network. We have designed three novel 3-phase handshake gossiping protocols, the DBHG, VBHG and AVBHG. These protocols can communicate much more effectively than the Gossiping and Adaptive Gossiping by producing significantly less duplicates, furthermore in the single message communication it can provide a faster coverage of the nodes then the other protocols.

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